MORSE CODE • OPERATING • REFERENCE

CW:

A Practical Reference

Codes, phrases, and operating procedures.



Valeriy Bagrintsev

English community edition v1.0.0 • 2026

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Codes, phrases, and operating procedures.

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A practical introduction to amateur Morse operation, with a reference section for operators at any level. Use it offline with a hand key and a notebook, alongside a radio, or with optional digital tools.

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This is an independently prepared guide. National rules, band plans, equipment manuals, and event regulations govern your actual operation. Examples use placeholder call signs and invented data.

AI tools were used in drafting, research support, layout, and consistency checks. Community corrections are welcome; see chapter 23.

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How to use this guide

A first Morse contact can be quite small: two call signs, a signal report, perhaps a name, and a friendly goodbye. Once you've done that, the same few building blocks will carry you through a surprising amount of radio.

This guide gives you both a way in and something to keep beside the key. Read the operating chapters when you're learning. Open the tables when you need a word, a signal, or a reminder. Experienced operators can go straight to the abbreviations, Q signals, phrase bank, and final desk card.

Choose your route

What you want to do	Start here
Make a first contact	Quick start (p. 7), then Your first contact (p. 19)
Learn by ear and by hand	Rhythm (p. 9), then Keys and a simple station (p. 11)
Look up something you heard	Abbreviations (p. 33), Q signals (p. 44), or Procedure signals (p. 17)
Practice without transmitting	Practice (p. 72) and Using the web (p. 74)
Keep a reminder within reach	The desk card (p. 87)

Reading the examples

`YOURCALL` means your authorized call sign. `THEIRCALL` means the other station's call sign. Replace the whole word; neither is sent literally. Example names, locations, reports, power levels, and frequencies are just that: examples.

[AR], [BT], and [SK] are joined procedure signals, often called **prosigns**. The brackets tell you to send one continuous character. You don't send the brackets. Ordinary groups such as CQ, DE, and TNX retain their normal letter spacing.

Dots and hyphens show the sound pattern. A space separates letters in written code. A slash between words in an exercise can stand for a word gap; a real slash inside a call sign is a character of its own, `-...-..`

Examples use international Morse and plain English. You can complete a routine contact with a small vocabulary. You can also send full words whenever they make you easier to understand.

A worldwide starting point

National rules decide where you may transmit, how you identify, and what your operating privileges allow. Band plans and event rules add their own practical details. Check the rules that apply where the transmitter is located. Listening, practicing with an oscillator, and learning on the web give you useful things to do before you have permission to transmit.

The operating advice here is aimed at ordinary amateur contacts. Formal message handling, emergency services, and specialized clubs may use additional procedures. When you join an organized activity, listen to its exchange and read its rules.

CHAPTER 01

Quick start: one contact at a time

Get ready

Have your call sign, a clock, and somewhere to write. Paper is fine. Set a sending speed at which you could understand an answer. Before transmitting, check the band, operating mode, antenna connection, and permitted power.

Find someone calling CQ at a comfortable speed. Listen long enough to get their complete call sign and learn what kind of contact they want. A relaxed CQ is a good first choice. A fast contest exchange is a different conversation.

Answer a call

When the caller finishes and listens, send:

```
THEIRCALL DE YOURCALL YOURCALL K
```

Now listen. If the answer contains your call sign, you're in. If it names someone else, give that station room to finish.

Start your own call

Listen on a suitable clear frequency. Ask:

```
QRL?
```

Wait for an answer. Ask once more if you're unsure. If someone says the frequency is in use, move. If you've heard no activity or reply, try a short call:

```
CQ CQ DE YOURCALL YOURCALL K
```

Leave a real listening gap. Another operator needs time to tune, recognize the call, and answer.

Give the essentials

```
THEIRCALL DE YOURCALL GE [BT] UR RST 559 559 [BT] NAME
ALEX ALEX [BT] QTH RIVERDALE RIVERDALE [BT] HW? K
```

That says: good evening; your report is 559; my name is Alex; my location is Riverdale; how do you copy? Substitute your own name and location, and the report you actually mean to give.

Four requests worth learning early

Send	You mean
UR CALL AGN PSE	Please repeat your call sign
NAME AGN PSE	Please repeat your name
MY RST AGN PSE	Please repeat the report you gave me
PSE QRS	Please send more slowly

Finish and write it down

```
TNX FER QSO 73 THEIRCALL DE YOURCALL [SK]
```

Record the other call sign, UTC date and time, band or frequency, CW, and both reports. Name and location are useful extras. A missing weather report isn't a problem; an uncertain call sign is worth asking about while the other station is still there.

CHAPTER 02

Hear the rhythm

Morse is a set of sound patterns with carefully placed silences. Those silences tell you where a character ends and a new word begins. Even a slow contact becomes easier when the spacing is steady.

The timing underneath it

Element	Standard length
Dot, or dit	1 unit
Dash, or dah	3 units
Gap inside a character	1 unit
Gap between characters	3 units
Gap between words	7 units

A dash is one longer sound. Three separate dots are a different character. The same distinction applies to silence: a letter gap isn't just an optional hesitation.

Use the printed patterns to check a character, then spend time hearing it. If you're counting every dot and dash, reduce the amount of text or give yourself longer gaps. Try recognizing the whole shape of a letter instead.

Character speed and text speed

Words per minute, or WPM, is commonly based on the reference word PARIS. One standard word occupies 50 timing units, including the final word gap. At standard spacing, a dot lasts `1200 / WPM` milliseconds. At 20 WPM, that's 60 milliseconds.

With **Farnsworth spacing**, the characters keep a brisk, recognizable shape while the gaps between characters and words grow longer. A trainer set to 20 WPM characters and 10 WPM effective speed gives your ear a clear sound and your brain more thinking time. Trainers label these settings differently, so check which value changes the characters and which changes the gaps.

Koch practice starts with a small character set and adds characters as your recognition improves. Koch and Farnsworth describe different choices; you can use them together.

Keep moving when you miss a character

Leave a blank and catch the next one. It feels untidy at first, but chasing one lost letter often costs you the next five. During a contact, a short request for the missing field is more useful than trying to reconstruct a whole paragraph.

As familiar groups become recognizable, let them stay together: CQ, DE, NAME, TNX. You may eventually hear a word without consciously spelling it. There's no deadline for that. Writing things down remains a useful operating skill.

Listen to your own sending

A recording is a patient practice partner. Send a few short words, stop, and play them back. Check that the letters have room to breathe. On a hand key, listen for dashes that shrink as your hand tires, or spaces that disappear near the end of a word.

Raise the speed when the present speed feels settled. A clean exchange at a modest pace is a contact you can enjoy.

CHAPTER 03

Keys, sound, and a simple station

You can learn a great deal with a key and a practice oscillator. The oscillator turns a contact closure into a local tone; no radio signal needs to leave the room. A transceiver adds the receiver, transmitter, and antenna needed for an on-air contact.

What each type of key does

Tool	What your hand controls
Straight key	The duration of each dot, dash, and space
Sideswiper, or cootie	Manual characters made with sideways movements
Mechanical bug	Automatic mechanical dots; manually timed dashes
Paddle with an electronic keyer	A keyer makes the dot or dash timing when you operate the paddle
Keyboard or memory sender	Stored or typed text becomes keyed characters

A paddle and a keyer are different things, even when the keyer is built into the radio. A dual-lever paddle also doesn't tell you which keying mode is selected. Iambic modes A and B differ in what happens when a squeeze is released; try them on a local tone and use the documented setting that feels comfortable.

For a straight key, adjust the contact gap and return force so that you can send without gripping or straining. Rest your

arm comfortably and take breaks. With a bug, make sure the mechanical dots are stable before adding hand-made dashes. Choose manageable practice text rather than forcing speed with tension.

Connect by function, not by plug size

A KEY socket, a microphone socket, and an audio output can all use similar-looking plugs. Their wiring and electrical limits may be completely different. Check the manuals for both devices before connecting them. An older transmitter may place voltages on the keying circuit that a modern electronic keyer cannot handle directly.

Test the local tone with transmitting disabled first. Check straight-key versus paddle mode, contact polarity where relevant, and the intended tip/ring connections. A stuck carrier is often a wiring or mode problem, not a reason to keep experimenting on a busy frequency.

Set up the receiver

Start with a comfortable audio pitch and a filter broad enough to find the signal. Once tuned, narrow the filter if it helps separate nearby stations. A very narrow filter can also hide an off-frequency reply or cut off a drifting signal.

The tone you hear is an audio difference produced by the receiver. A 600 Hz listening tone isn't a 600 Hz radio frequency. Use the radio's documented CW tuning indication or receive offset so that your transmitted signal lands where the other operator expects it.

A sidetone lets you hear your own sending. Set it to a comfortable level. It should help your timing without making you miss the return to receive.

On-air CW and practice tones

In ordinary CW radio operation, the transmitter keys its carrier on and off. Feeding an audio tone into an FM voice channel is another way of carrying Morse information, but the receiving mode and occupied signal are different. A practice oscillator, an internet Morse room, and a CW radio contact can all teach timing; keep track of which kind of connection you're making.

A small offline kit

A hand key, oscillator, pencil, clock, and a page of exercises make a useful starting kit. Add a receiver when you want real signals. Add software when it solves a particular problem for you. None of the exercises in this guide depends on a particular brand of radio or decoder.

CHAPTER 04

Letters, numbers, and punctuation

International letters

Letter	Pattern	Letter	Pattern
A	.-	N	-.
B	-...	O	---
C	-.-.	P	.--.
D	-..	Q	--.-
E	.	R	.-.
F	..-.	S	...
G	--.	T	-
H	....	U	..-
I	..	V	...-
J	.---	W	.--
K	-.-	X	-..-
L	.-..	Y	-.--
M	--	Z	--..

Numbers

Digit	Pattern	Digit	Pattern
0	`-----`	5	`.....`
1	`.----`	6	`-....`
2	`..---`	7	`--...`
3	`...--`	8	`---..`
4	`....-`	9	`----.`

Punctuation you may actually use

Mark	Pattern	Use
.	`.-.-.-`	Full stop
,	`--..--`	Comma
?	`..--..`	Question or request for a repeat
/	`-..-.`	Slash, including within call signs
-	`-....-`	Hyphen or minus
=	`-...-`	Double hyphen; also the [BT] pattern
+	`.-.-.`	Plus; also the [AR] pattern
:	`---...`	Colon
'	`.----.`	Apostrophe
"	`.-..-.`	Quotation mark
(	`-.--.`	Opening parenthesis
)	`-.--.-`	Closing parenthesis
@	`.--.-.`	Email at-sign

Punctuation is sparse in many radio contacts. Short blocks, sensible pauses, and [BT] often do most of the work. The question mark is especially useful, but a named request such as NAME AGN PSE is clearer than a question mark by itself.

Some programs support extra symbols or national characters. Confirm the alphabet at both ends before relying on them. International call signs should be entered into a log with their proper Latin letters and digits, including any relevant slash suffix.

Cut numbers

In a numeric exchange, you may hear N for 9 and T for 0. A sometimes stands for 1. Thus **5NN** is a shortened 599. These are conventions used in context, not replacements for the normal alphabet.

Send full digits while you're learning. Don't substitute cut numbers into a call sign. If you're unsure whether you heard a letter or a shortened number, ask for the field again.

Useful listening contrasts

Practice A/N, U/D, V/B, K/R, and F/L in mixed groups. For E/I/S/H/5, pay attention to the length and rhythm of the whole character. For T/M/O/0, do the same with dashes. Mix the groups often enough that you must listen rather than remember their order.

CHAPTER 05

Procedure signals: who sends next

Procedure signals make the conversation easy to follow. Learn a few well and use the rest as you encounter them.

Signal	Pattern	Meaning
K	-.-	Your turn; an invitation to transmit
[KN]	-.--.	The named station is invited to reply
[AR]	.-.-.	End of a message
[SK]	...-.-	End of the contact
[BT]	-...-	A separator between thoughts
[AS]	.-...	Wait
[VE]	...-.	Understood, in procedural use
[HH]	....	An error: eight dots
[SOS]	...---...	Distress signal; not an on-air practice call

K, [AR], and [SK] do different jobs

Finish a general call with K so that someone knows you're listening. During a contact, K hands the conversation back. [KN] makes the intended recipient explicit.

[AR] marks the end of a message. [SK] marks the end of the contact. You don't need to string every signal together at the end of every transmission. For a first contact, a clear call sign exchange, K, and a clean closing will take you a long way.

You will see joined characters printed with a bar above them, in angle brackets, or with no special formatting. This guide uses square brackets consistently. KN is also heard as separate letters. The joined [KN] pattern is identical to an opening parenthesis; its position in the exchange tells you which meaning is intended.

BK and CL

BK normally has a letter gap: `-... -.-`. It can mark a short handover once both operators know who's talking. Some operators join it, so expect variation when listening.

CL is sent as ordinary letters and means the station is closing. A station that signs off with CL may not be listening for another call. Give them the chance to leave.

Fix a sending error

Stop, give [HH] if useful, and repeat the word or group from the beginning. For a small obvious slip, a pause and a clean repeat may be enough. Repeat a mistaken call sign in full rather than sending one unexplained letter.

The familiar two E characters sometimes exchanged after goodbye are an informal farewell. They aren't required to complete a contact. In a biography or club notice, SK may instead mean Silent Key, a radio amateur who has died; that is a different use of the letters.

Keep ordinary letters separate

CQ, DE, TNX, TU, and 73 are regular groups. Their familiarity doesn't remove the spaces between their characters. Joined prosigns are the exceptions you deliberately learn.

CHAPTER 06

Your first contact, with a worked example

Pick an easy conversation

Choose a signal you can hear steadily and a speed you can follow. Listen through a complete exchange if possible. You'll learn the caller's rhythm and find out whether they're chatting, working an event, or handling a queue.

Write down their full call sign before you call. If you use a decoder, compare its text with what you hear. A plausible-looking call sign on a screen can still be wrong.

The exchange

In this example, **A** is THEIRCALL, a station calling CQ. **B** is you, YOURCALL. Sam and Alex are example names. The reports differ on purpose.

1. A calls

```
CQ CQ DE THEIRCALL THEIRCALL K
```

A has asked for a contact and switched to receive.

2. B answers

```
THEIRCALL DE YOURCALL YOURCALL K
```

This makes the addressee clear and gives A a second chance to copy your call.

3. A gives the first information

```
YOURCALL DE THEIRCALL GE [BT] UR RST 579 579 [BT] NAME
SAM SAM [BT] QTH HILLTOWN HILLTOWN [BT] HW? [KN]
```

A has given **you 579**, introduced Sam, and sent a location. Write down the parts you need. If you missed the name, ask for it; there's no need to abandon the contact.

4. B replies

```
THEIRCALL DE YOURCALL R TNX SAM [BT] UR RST 559 559 [BT]
NAME ALEX ALEX [BT] QTH RIVERDALE RIVERDALE [BT] HW? [KN]
```

You give **A 559**, then your own name and location. R says that the preceding information was received. If only some of it was received, say what you copied and ask about the missing part instead of confirming everything.

5. A continues

```
R ALEX [BT] HR PWR 10 W ANT DIPOLE [BT] WX CLOUDY [BT]
[KN]
```

This is an invitation to a little conversation. You can describe your station, ask a question, or politely finish. A weather report is optional.

6. B closes

```
R SAM [BT] TNX FER QSO ES UR PATIENCE [BT] 73 THEIRCALL
DE YOURCALL [SK]
```

You're thanking Sam and ending the contact.

7. A says goodbye

```
TU ALEX 73 YOURCALL DE THEIRCALL [SK]
```

That's the exchange. There's no need to add another long transmission.

The log entry

Field	In this example
Other station	THEIRCALL, replaced with the actual call sign
Report sent	559: your assessment of A's signal
Report received	579: A's assessment of your signal
Name and location	Sam, Hilltown
Date, time, frequency	The actual UTC values and operating frequency
Mode	CW

The names and towns here are invented training data. Don't copy them into a real log. You can shorten the conversation considerably once you understand what each block does.

CHAPTER 07

Calling, answering, and having a conversation

A short CQ leaves space for an answer

A useful call gives people enough time to notice you without making them wait through a long announcement. CQ two or three times, your call sign a couple of times, then K is a reasonable place to start. Conditions may justify a little more repetition.

After the call, listen for a reply that may be slightly off your receive frequency. An extremely narrow filter can make an answer easy to miss. Don't keep restarting CQ so quickly that a slower operator never gets in.

When you answer

For a relaxed contact, THEIRCALL DE YOURCALL K identifies both stations clearly. In a busy DX queue or contest, the normal answer may be just your complete call sign once. Match the situation you've been listening to.

Listen for the answer addressed to you. If it names another station, stay quiet. If the operator is trying to complete a partial call that clearly isn't yours, help by leaving them alone.

Choosing a topic

Once you've exchanged names and reports, pick one thing to talk about. An antenna someone built, a portable outing, a new key, or a recent repair can become a good conversation.

```
I BUILT A DIPOLE THIS WEEK [BT] WHAT ANT ARE U USING?
[KN]
```

Keep the first question short. Give the other person room to respond. You can always follow up once you know how well they are copying.

A long first transmission containing every detail of your station can be hard work for a newcomer. One or two thoughts at a time usually makes for a more enjoyable exchange.

A quick handover

After the contact is established, short BK exchanges can feel quite natural:

A: `BK HR PWR 5 W BK`
B: `BK FB HR 10 W BK`

Continue to identify as required by your local rules. Return to full call signs if there is confusion or another operator joins.

Calling after another contact

Wait until both stations have finished. Then call the station you want. A clear closing doesn't always mean the operator wants another contact; CL is a particularly good hint that they're leaving.

To join a conversation already in progress, wait for a suitable gap and give your call briefly. Then wait for an invitation. In a roundtable, make the next recipient clear so that several people don't all start together.

Directed calls

A directed CQ asks for a particular region, group, or type of contact. Respect it. CQ DX generally seeks distant stations, but what counts as DX depends on the band, location, and caller's intention. It isn't a universal distance or a simple border crossing.

CHAPTER 08

Repeats, slow sending, and difficult signals

A request for a repeat is part of a normal contact. The most useful request tells the other operator exactly what you need.

What happened	A useful response
You missed the call sign	`UR CALL AGN PSE`
You copied the report but lost the name	`RST CPY [BT] NAME AGN PSE`
The sending is too fast	`PSE QRS`
You need a slower, stated speed	`PSE QRS 10 WPM`
Letters are clear but words run together	`PSE MORE SPACE BETWEEN WORDS`
You missed the report they gave you	`MY RST AGN PSE`
A word disappeared in fading	`LAST WORD AGN PSE`
You need a short pause	`[AS]`
They copied your call incorrectly	Repeat your complete call sign clearly
The contact is no longer readable	`SRI NIL CPY TNX 73 [SK]`, if you can send

Acknowledge only what you have

R ALL OK is a strong statement. If you have the report but not the name, say so. That lets the other operator repeat one field instead of guessing where you became lost.

If someone asks you for NAME AGN, send the name clearly, perhaps twice. Increasing the speed or repeating the entire preceding transmission makes their job harder.

When the receiver is struggling

Check the tuning before changing everything else. Try a filter that reduces nearby signals without cutting off the one you want. Set comfortable gain and volume. Very loud audio may make listening tiring and can overload a decoder input.

Fading, interference, uneven hand sending, and two stations on top of each other can all confuse decoding software. Treat the decoded line as a clue. Ask for an uncertain call sign or report again.

When nobody answers

Check the transmit frequency, antenna connection, mode, and power setting. Check whether the station is operating split. Then try another suitable caller or another time. Hearing someone well doesn't guarantee they can hear you equally well.

A spot on a website is useful for finding a station, but it doesn't prove that the station is still there or listening where you expect. The current on-air instructions matter most.

When a station disappears

Listen for a while, then try a brief addressed call if appropriate. If there's no return, end the attempt. Mark uncertain or incomplete data in your notes. An earlier conversation, a callbook entry, or a cluster listing can't supply words you didn't receive.

CHAPTER 09

Good manners on a shared band

The best courtesy is often simply leaving space: space between words, space for a reply, and enough room for someone else's contact.

Before occupying a frequency

Listen, then ask QRL?. You may hear only one side of an existing contact. A quiet moment doesn't necessarily mean the frequency is free. If someone replies that it is occupied, find another spot without starting a debate.

Avoid tuning up over a conversation. Use a suitable dummy load for tests that don't need an antenna, or follow your radio's documented short tuning procedure on a clear, permitted frequency at an appropriate power.

Make the other operator's job easier

Send a complete call sign clearly. Leave useful pauses. Slow down when asked. Keep your blocks short if the other person is working hard to copy. A little patience is more memorable than a fast exchange full of uncertainty.

If you need to stop, say so and close. If the other operator has a queue or limited battery time, let a short contact stay short. Neither of you owes the other a long conversation.

Treat mistakes as mistakes first

Someone may have the wrong VFO selected, miss a directed call, or misunderstand the turn-taking. A brief courteous correction can help. Adding an angry carrier, repeated dots, or insults only makes another signal for everyone to fight through.

Before asking someone to move, check your own receiver. If a nearby signal makes the contact difficult and there is room elsewhere, moving may be the easiest answer.

In a pileup

Send your call once and listen. Let the called station finish with the person they selected. Use your whole call, rather than a suffix or a power category. Follow requests for a particular area or partial call. The operator handling the queue needs a chance to hear an answer.

A scheduled contact is still on a shared frequency

An arrangement to meet, or sked, is a plan rather than a reservation. If the frequency is busy, use an agreed alternative or find a clear one. A website spot also doesn't confer ownership.

If you hear a newcomer

Try a shorter sentence, a little extra word spacing, or a slower reply. Repeat the important field without fuss. You can help someone have a successful contact without turning it into an examination.

Real distress

Distress signals deserve attention and should never be used as casual on-air exercises. If you hear an apparent immediate threat to life, avoid causing interference, copy the location and message, note the time, and follow the emergency procedures available in that situation. Where an official emergency service can be reached directly, use it. Amateur radio is one possible communication path, not a guarantee of a response.

CHAPTER 10

Reports, numbers, frequencies, and time

A signal report describes the signal you are receiving. It isn't a measurement of the other transmitter's advertised power, and it doesn't have to match the report you receive.

RST at a glance

Part	Range	What you're judging
R: readability	1-5	How well you can make out the message
S: strength	1-9	The received signal's strength
T: tone	1-9	The quality of the keyed tone

Readability

R	Practical meaning
1	Not readable
2	Only occasional characters or words get through
3	Readable with substantial effort
4	Readable with little difficulty
5	Fully readable

Strength

S	Practical meaning
1	Barely perceptible
2	Very weak
3	Weak
4	Moderate
5	Fairly good
6	Good
7	Fairly strong
8	Strong
9	Very strong

Tone

The historical T scale describes roughness and ripple in the tone. It runs from a very rough, broad note at T1 to a clean tone at T9. T2-T4 describe increasingly less rough notes; T5-T6 still have obvious ripple; T7-T8 approach a clean note with progressively less residual modulation.

A weak signal can still have a clean tone. Many sound modern transmitters merit T9, even when their signals are faint. If the issue is a chirp or key clicks, say what you hear rather than inventing an unfamiliar number.

Report	How to read it
599	Fully readable, very strong, clean tone
579	Fully readable, fairly strong, clean tone
559	Fully readable, strength 5, clean tone

Report	How to read it
339	Readable with effort, weak, clean tone
5NN	A shortened 599 in context

An S-meter can help, but readings depend on calibration and receiver settings. Use a report that communicates something useful. In many contests, 599 functions as a conventional part of the exchange; in a relaxed contact, an honest report is more informative.

Other scales are different

QSA uses a formal strength scale of 1-5. QRI has a formal tone scale of 1-3: good, variable, bad. Neither should be silently substituted for S or T in an RST report. For an ordinary CW contact, RST is usually the simpler choice.

Keep units clear

A frequency written as 14060 normally means 14060 kHz, or 14.060 MHz, in the relevant HF context. Include KHZ or MHZ when it could be misunderstood. An audio pitch such as 600 Hz is a separate receiver setting.

PWR 5 W reports five watts. ABT 5 W says approximately five watts. Describe the power you know or have selected; don't turn a product label into a claimed measurement.

Use UTC for scheduled contacts and logs unless you explicitly agree otherwise. A clock showing UTC avoids a lot of arithmetic. When converting from local time, check the date too: near midnight, the UTC date can differ from the local date.

Temperatures need a unit when it matters. TEMP PLUS 12 C and TEMP 54 F are clear. Distances and antenna heights also benefit from KM, M, or another explicitly stated unit.

CHAPTER 11

Abbreviations A-F

Look up the group you heard, then read the example in context. These are shared habits, not a requirement to abbreviate every word. Full words are welcome when they are clearer.

The context does some of the work

RPT can ask for a repeat or refer to a report. FM may mean from in a sentence and frequency modulation in a technical discussion. R acknowledges reception; C or YES is a clearer affirmative answer to a question. If the intended meaning is uncertain, ask or use a full word.

Reference table 1

Group	Meaning	Example or note
AA	All after	PSE AGN AA NAME
AB	All before	PSE AGN AB QTH
ABT	About; approximately	ABT 10 W
ADR / ADDR	Address	ADR VIA CALLBOOK
AF	Audio frequency	AF GAIN
AGN	Again; repeat	NAME AGN PSE
ANR	Another	ANR CONTACT
ANS	Answer	TNX FER ANS
ANT	Antenna	ANT DIPOLE
AR	See joined [AR] in the procedure table	End of a message

Group	Meaning	Example or note
ARND	Around; approximately	ARND 1500 UTC
AS	See joined [AS] in the procedure table	Wait
B4	Before	WKD U B4
BCI	Broadcast interference	HV BCI
BCNU	Be seeing you	TNX BCNU
BK	Break; a short handover, by context	BK HW? BK
BT	See joined [BT] in the procedure table	Separate two thoughts
BTR	Better	NW BTR CPY
BTU	Back to you	BTU K
BUG	Mechanical semiautomatic key	USING A BUG

Reference table 2

Group	Meaning	Example or note
BURO	QSL bureau	QSL VIA BURO
C	Yes; correct	C UR CALL OK
CALL / CS	Call sign	UR CALL AGN
CBA	Callbook address	QSL TO CBA
CFM	Confirm; I confirm	CFM UR NAME
CK	Check	PSE CK
CL	Closing the station	QRT NW CL
CLDY	Cloudy	WX CLDY

Group	Meaning	Example or note
CLG	Calling	STN CLG AGN PSE
CONDX / CX	Conditions	CONDX POOR
CPY / CPI	Copy; receive the message	HW CPY?
CQ	A general call	CQ CQ DE YOURCALL K
CU	See you	CU SN
CUAGN	See you again	HPE CUAGN
CUD	Could	CUD U QRS?
CUL	See you later	73 CUL
CUZ / COS	Because	QRT CUZ QRM
CW	Continuous wave; usual shorthand for Morse radio operation	WORKING CW
DE	From; this is	THEIRCALL DE YOURCALL
DN	Down	DN 1, when a frequency offset is intended

Reference table 3

Group	Meaning	Example or note
DR	Dear	GE DR SAM
DX	Distant contact or station, according to context	CQ DX
ES	And	73 ES GL
FB	Fine; excellent	FB SIG

Group	Meaning	Example or note
FER	For	TNX FER QSO
FM	From; also frequency modulation in a technical context	FM HOME
FREQ	Frequency	FREQ AGN PSE
FWD	Forward	FWD MSG

CHAPTER 12

Abbreviations G-P

Look up the group you heard, then read the example in context. These are shared habits, not a requirement to abbreviate every word. Full words are welcome when they are clearer.

Friendly language travels better

OM and YL are traditional operator terms. A person's name or a simple greeting is always an option. You don't need to reproduce every expression you hear. In particular, recognizing an insulting term doesn't oblige you to use it.

Reference table 1

Group	Meaning	Example or note
GA	Good afternoon; go ahead in another context	GA ALEX
GD / GUD	Good	GUD SIG
GE	Good evening	GE SAM
GL	Good luck	73 ES GL
GLD	Glad	GLD TO MEET U
GM	Good morning	GM OM
GN	Good night	GN ES 73
GND	Ground	GND CONNECTION
GP	Ground-plane antenna, in an antenna discussion	ANT GP

Group	Meaning	Example or note
HI / HI HI	Laughter; a light-hearted remark	RAIN AGN HI HI
HNY	Happy New Year	HNY ES 73
HPE	Hope	HPE CUAGN
HR	Here; hear in some contexts	HR PWR 5 W
HRD	Heard	HRD U B4
HRS	Hours	TWO HRS
HV	Have	HV QRM
HW	How; how do you copy?	HW CPY?
INFO	Information	TNX FER INFO
K	Invitation to transmit	YOURCALL K
KN	See [KN]; also heard as separate letters	The named station should answer

Reference table 2

Group	Meaning	Example or note
LID	An unkind label for a poor operator	Recognize it; avoid using it as an insult
LP	Long path, in propagation discussions	VIA LP
LSN	Listen	LSN UP
LTR	Later; letter in another context	CU LTR
M / MTR	Meter, by context	ANT UP 8 M

Group	Meaning	Example or note
MIN	Minute	QRX 1 MIN
MNI	Many	MNI TNX
MSG	Message	MSG AGN PSE
N	No; also a cut 9 in a numeric field	N, or 5NN in a report
NCS	Net control station	NCS DE YOURCALL
NIL	Nothing; no copy by context	SRI NIL CPY
NR	Number; near in another context	NR 007
NW	Now	QRT NW
OK	All right; correct	ALL OK
OM	Traditional friendly address to a male operator	TNX OM
OP / OPR	Operator	OP ALEX
OT	Old-timer; experienced operator	OP OT
PA	Power amplifier	PA OFF
PSE / PLS	Please	PSE QRS
PWR	Power	PWR 10 W

CHAPTER 13

Abbreviations R-Z and number greetings

Look up the group you heard, then read the example in context. These are shared habits, not a requirement to abbreviate every word. Full words are welcome when they are clearer.

Number greetings are conventions

73 is the useful everyday closing. Some groups use 72, 88, or other numbers with a particular social meaning. They aren't universal commands. SK in a notice about a person is separate from the joined [SK] that ends a contact.

Reference table 1

Group	Meaning	Example or note
R	Received; other meanings depend on context	R TNX
RCVR / RX	Receiver; receive	RX OK
RFI	Radio-frequency interference	LOCAL RFI
RIG	Radio equipment	RIG HOMEBREW
RPRT	Report	TNX FER RPRT
RPT	Repeat; report in another context	RPT NAME PSE
RST	Readability, strength, tone report	UR RST 579

Group	Meaning	Example or note
RTTY	Radioteletype	ALSO WORK RTTY
SASE	Self-addressed stamped envelope	Follow the QSL recipient’s instructions
SIG / SIGS	Signal or signals; signature in message handling	UR SIGS QSB
SK	See [SK]; Silent Key in a biographical context	End the contact; a different meaning in a club notice
SKED	An arranged contact	SKED AT 1400 UTC
SN	Soon	CU SN
SNR	Signal-to-noise ratio	SNR IMPROVED
SP	Short path, in propagation discussions	VIA SP
SRI	Sorry	SRI NAME NIL
SSB	Single sideband	ALSO WORK SSB
STN	Station	STN CLG PSE AGN
T	A cut zero in a numeric field	TT1 may mean 001
TEMP	Temperature	TEMP PLUS 12 C

Reference table 2

Group	Meaning	Example or note
TFC	Traffic; messages	NO TFC
TKS / TNX	Thanks	TNX FER CALL
TMW	Tomorrow	QRV TMW?

Group	Meaning	Example or note
TU	Thank you	TU 73
TX / XMTR	Transmitter; transmit	TX PWR 5 W
U	You	TNX U
UFB	Very good; emphatic FB	UFB CPY
UP	Up; higher, depending on context	UP 1 in a split-frequency call
UR / YR	Your; you are	UR RST 559
URS	Yours	ANT SIMILAR TO URS
UTC / Z	Coordinated Universal Time	AT 1200 UTC
VFO	Variable-frequency oscillator; a radio's tuning channel	CHECK TX VFO
VY	Very	VY FB
W	Watts, when used as a unit	5 W
WA	Word after	PSE AGN WA NAME
WB	Word before	PSE AGN WB QTH
WDS	Words	LAST WDS AGN
WID	With	QSO WID U
WKD	Worked; made contact	WKD U B4
WKG	Working	WKG PORTABLE

Reference table 3

Group	Meaning	Example or note
WL	Will	WL CALL TMW
WPM	Words per minute	PSE QRS 10 WPM
WUD	Would	WUD U QSY?
WX	Weather	WX SUNNY
XCVR	Transceiver	XCVR QRP
XTAL	Crystal	XTAL CONTROLLED
XYL	Traditional term for wife	Understand the term; a name is often more natural
YL	Traditional address or description for a female operator	An operator term, not an age test
73	Best regards	TNX QSO 73
72	A friendly greeting used in some QRP circles	72 ES GL
88	Love and kisses; a more personal greeting	Use when appropriate to the relationship

CHAPTER 14

Q signals: questions and everyday use

Q signals grew out of formal radio procedures. Amateur operators use a small subset, often as everyday shorthand. A question mark generally makes the group a question, but removing it does not always mean yes. Each code has its own meaning.

The entries below separate the formal question and reply from modern conversational use. You can understand both without making every friendly contact sound like a formal telegram.

QRA

Form	Meaning
With ?	What is the name of your station?
Reply or instruction	The station name is ...
On the amateur bands	For an operator's personal name, NAME is usually clearer.

QRB

Form	Meaning
With ?	How far apart are our stations?
Reply or instruction	Our approximate distance is ...
On the amateur bands	Give a unit, such as KM.

QRG

Form	Meaning
With ?	Will you tell me my exact frequency?
Reply or instruction	Your exact frequency is ...
On the amateur bands	QRG 14060 KHZ

QRH

Form	Meaning
With ?	Does my frequency vary?
Reply or instruction	Your frequency varies.
On the amateur bands	A drift report; describe what you hear.

QRI

Form	Meaning
With ?	How is the tone of my transmission?
Reply or instruction	Tone: 1 good, 2 variable, 3 bad.
On the amateur bands	This formal scale is not T1-T9 in RST.

QRK

Form	Meaning
With ?	How intelligible is my signal?
Reply or instruction	Intelligibility is ... on a 1-5 scale.
On the amateur bands	Most casual contacts use RST instead.

QRL

Form	Meaning
With ?	Are you busy?
Reply or instruction	I am busy; please do not interfere.
On the amateur bands	QRL? is the usual check before occupying a frequency.

QRM

Form	Meaning
With ?	Is my transmission being interfered with?
Reply or instruction	Your transmission is being interfered with.
On the amateur bands	In ordinary ham use, SRI QRM usually describes interference at my receiver.

QRN

Form	Meaning
With ?	Are you troubled by static?
Reply or instruction	I am troubled by static.
On the amateur bands	Commonly used for atmospheric noise; describe local electrical noise more specifically.

QRO

Form	Meaning
With ?	Shall I increase transmitter power?
Reply or instruction	Increase transmitter power.
On the amateur bands	HR QRO is a separate conversational use meaning higher-power operation.

QRP

Form	Meaning
With ?	Shall I decrease transmitter power?
Reply or instruction	Decrease transmitter power.
On the amateur bands	HR QRP means low-power operation; give watts when the value matters.

QRQ

Form	Meaning
With ?	Shall I send faster?
Reply or instruction	Send faster.
On the amateur bands	PSE QRQ

QRS

Form	Meaning
With ?	Shall I send more slowly?
Reply or instruction	Send more slowly.
On the amateur bands	PSE QRS 10 WPM

QRT

Form	Meaning
With ?	Shall I stop sending?
Reply or instruction	Stop sending.
On the amateur bands	QRT NW is commonly used to say I am closing down now.

QRU

Form	Meaning
With ?	Have you anything for me?
Reply or instruction	I have nothing for you.
On the amateur bands	Without the question mark, this is a negative answer.

QRV

Form	Meaning
With ?	Are you ready?
Reply or instruction	I am ready.
On the amateur bands	QRV AT 1500 UTC

QRW

Form	Meaning
With ?	Shall I tell another station you are calling?
Reply or instruction	Please tell that station I am calling.
On the amateur bands	Name the station and frequency as needed.

QRX

Form	Meaning
With ?	When will you call again?
Reply or instruction	I will call again at the stated time and frequency.
On the amateur bands	QRX 1 MIN is commonly used as a request to wait.

QRY

Form	Meaning
With ?	What is my turn?
Reply or instruction	Your turn is number ...
On the amateur bands	Relates to a communication queue; uncommon in a casual contact.

QRZ

Form	Meaning
With ?	Who is calling me?
Reply or instruction	You are being called by ...
On the amateur bands	Use after a call you need to identify; not as a clear-frequency check.

QSA

Form	Meaning
With ?	What is the strength of my signal?
Reply or instruction	Strength is ... on the formal 1-5 scale.
On the amateur bands	Not the S1-S9 scale in RST.

QSB

Form	Meaning
With ?	Are my signals fading?
Reply or instruction	Your signals are fading.
On the amateur bands	UR SIGS QSB

QSD

Form	Meaning
With ?	Are my signals mutilated?
Reply or instruction	Your signals are mutilated.
On the amateur bands	In CW conversation it often refers to defective keying; ask tactfully.

QSK

Form	Meaning
With ?	Can you hear between your signals, and may I break in?
Reply or instruction	I can hear between my signals; you may break in.
On the amateur bands	Full break-in depends on the station’s capabilities and settings.

QSL

Form	Meaning
With ?	Can you acknowledge receipt?
Reply or instruction	I acknowledge receipt.
On the amateur bands	A spoken or keyed QSL does not itself create an electronic award confirmation.

QSN

Form	Meaning
With ?	Did you hear me or the named station on that frequency?
Reply or instruction	I heard the specified station there.
On the amateur bands	HRD U ON 20M is a plain alternative.

QSO

Form	Meaning
With ?	Can you communicate with the named station?
Reply or instruction	I can communicate with that station.
On the amateur bands	As a noun, QSO simply means a radio contact.

QSP

Form	Meaning
With ?	Will you relay a message?
Reply or instruction	I will relay it.
On the amateur bands	Identify the recipient and the message.

QSU

Form	Meaning
With ?	Shall I send or reply on the stated frequency?
Reply or instruction	Send or reply on that frequency.
On the amateur bands	QSY is more familiar for an ordinary change of frequency.

QSV

Form	Meaning
With ?	Shall I send a series of Vs?
Reply or instruction	Send a series of Vs.
On the amateur bands	A test request; avoid creating interference.

QSX

Form	Meaning
With ?	Will you listen to the named station on that frequency?
Reply or instruction	I am listening to that station on that frequency.
On the amateur bands	QSX followed by a frequency commonly announces where a station is listening.

QSY

Form	Meaning
With ?	Shall I change transmission to another frequency?
Reply or instruction	Change transmission to another frequency.
On the amateur bands	PSE QSY 14060; first make sure the new spot is usable.

QSZ

Form	Meaning
With ?	Shall I send each word or group more than once?
Reply or instruction	Send each word or group the specified number of times.
On the amateur bands	PSE QSZ 2

QTC

Form	Meaning
With ?	How many messages have you to send?
Reply or instruction	I have ... messages.
On the amateur bands	Some contests have a special QTC exchange; learn its rules first.

QTH

Form	Meaning
With ?	What is your position?
Reply or instruction	My position or location is ...
On the amateur bands	QTH RIVERDALE is normal conversational ham usage.

QTR

Form	Meaning
With ?	What is the correct time?
Reply or instruction	The correct time is ...
On the amateur bands	State UTC unless another basis is explicit.

QTU

Form	Meaning
With ?	What are your operating hours?
Reply or instruction	I operate between the stated hours.
On the amateur bands	An agreed operating schedule.

QUA

Form	Meaning
With ?	Have you news of the named station?
Reply or instruction	I have news of that station.
On the amateur bands	A less common code; full words may be easier.

The distinctions that save time

Groups	Keep the difference clear
QRL? / QRZ?	Is this in use? / Who called me?
QRS / AGN	Send slower / Send it again
QSY / QRT	Change frequency / Stop sending
QSB / QSD	Fading / A problem with the transmitted characters or keying
QSL / QSO	Acknowledgment / A contact
QSA / S in RST	A 1-5 strength scale / A 1-9 strength scale

CHAPTER 15

Phrase bank: find the line you need

Choose a line that does the job, change its data, and leave room for the reply. You can always use a full word instead of an abbreviation.

Finding room and calling

Send	Meaning
QRL?	Is this frequency in use?
CQ CQ DE YOURCALL YOURCALL K	A short general call
CQ QRS DE YOURCALL K	A call inviting a slower contact
CQ DX DE YOURCALL K	A call seeking distant stations
THEIRCALL DE YOURCALL K	An addressed call or a clear answer to CQ
YOURCALL YOURCALL	A brief answer when that fits the operating style
QRZ?	Who called me?
STN CLG AGN PSE	Calling station, please repeat
PSE SEND FULL CALL	Please give your complete call sign
UR CALL AGN PSE	Please repeat your call sign

Opening the contact

Send	Meaning
R TNX FER CALL	Received; thanks for calling
GM ALEX	Good morning, Alex
GA SAM	Good afternoon, Sam
GE DR OM	Good evening; a traditional friendly greeting
UR RST 559 559	Your report is 559
NAME SAM SAM	My name is Sam
OP ALEX	The operator is Alex
QTH RIVERDALE	My location is Riverdale
HW CPY? K	How do you copy? Your turn
R ALL OK	Everything was received

Asking for a repeat

Send	Meaning
UR NAME AGN PSE	Please repeat your name
MY RST AGN PSE	Repeat the report you gave me
QTH AGN PSE	Repeat your location
NR AGN PSE	Repeat the number
LAST WORD AGN PSE	Repeat the last word
PSE AGN WA NAME	Repeat the word after NAME
PSE AGN AA RST	Repeat everything after RST
PSE SEND NAME TWICE	Send the name twice

Send	Meaning
RST CPY NAME NIL	I copied the report, but not the name
SRI QUESTION AGN PSE	Sorry, please repeat the question

Slowing down and waiting

Send	Meaning
PSE QRS	Please send more slowly
PSE QRS 10 WPM	Please send about 10 words per minute
PSE MORE SPACE BETWEEN WORDS	Please leave longer word gaps
I AM NEW TO CW	I am a newcomer to Morse operation
THIS IS MY FIRST CW QSO	This is my first CW contact
MNI TNX FER UR PATIENCE	Thanks very much for your patience
[AS]	Please wait
QRX 1 MIN	Wait one minute, in common amateur usage
PSE START AGAIN FM NAME	Start again from the name
ONLY LAST TWO LETTERS CPY	I copied only the final two letters

Signal and station

Send	Meaning
SRI QRM	Interference is making reception difficult
HV QRN	I have atmospheric noise
LOCAL RFI HR	I have local radio-frequency interference
UR SIGS QSB	Your signals are fading
UR SIGS FB	Your signals are good
NW BTR CPY	Reception is better now
HR PWR 5 W	I am using five watts
ANT DIPOLE UP 8 M	A dipole eight meters high
ANT VERTICAL	A vertical antenna
I AM USING A STRAIGHT KEY	I am sending with a straight key
USING PADDLE ES KEYER	I am using a paddle and electronic keyer
RIG HOMEBREW	The radio is home-built
WHAT ANT ARE U USING?	Ask about the other antenna
PSE CHECK KEYING	A tactful request to check the sending
USING A DECODER	I am using a decoder

A little conversation

Send	Meaning
WX SUNNY	Sunny weather here
WX CLOUDY	Cloudy weather here
WX RAIN	It is raining here
TEMP PLUS 12 C	Temperature is plus twelve Celsius
I BUILT MY ANT	I built my antenna
I ENJOY PORTABLE OPERATION	I enjoy operating away from home
HOW LONG HAVE U BEEN ON CW?	Ask about experience
FIRST TIME ON THIS BAND	My first time on this band
QSO B4 TNX	We have worked before; thanks
GLD TO MEET U	Glad to meet you

Moving and making plans

Send	Meaning
PSE QSY 14060	Please move to 14060 kHz, if usable and permitted
QRV TMW AT 1500 UTC?	Will you be available tomorrow at 15:00 UTC?
SKED AT 1500 UTC	An arrangement to meet at 15:00 UTC
PSE CONFIRM TIME IN UTC	Please confirm the UTC time
I WILL CALL U AGAIN	I will call you again

Send	Meaning
PSE WAIT I CHECK MY ANT	Please wait while I check my antenna
UP 1	Listening one kHz above, in a split call
PSE CONFIRM FREQ	Please confirm the frequency

Closing and confirmations

Send	Meaning
TNX FER QSO	Thanks for the contact
TU 73	Thank you; best regards
HPE CUAGN	Hope to see you again
73 ES GL	Best regards and good luck
GN ES GUD DX	Good night and good distant contacts
MUST QRT NW	I need to close down now
SRI BATTERY LOW MUST QRT	Battery is low; I need to stop
TNX QSO 73 THEIRCALL DE YOURCALL [SK]	A complete closing
PSE QSL VIA BURO	Please send a QSL through the bureau
QSL INFO ON MY PAGE	My confirmation instructions are on my page

Events and brief exchanges

Send	Meaning
CQ TEST DE YOURCALL K	A contest call
5NN 007	A report and serial number, only when the rules call for them
NR 007 007	A repeated number
TU YOURCALL TEST	Thanks; continuing the contest
REF AGN PSE	Repeat the activity reference
PSE SEND UR CLUB NR	Please send your club membership number
I AM NOT A MEMBER	I do not have a club membership number
TNX FER FIRST QSO	Thanks for a first contact

CHAPTER 16

DX, queues, and split operation

DX operating can be a brief exchange with a distant station or a relaxed conversation across a long path. When many people call one station at once, the queue is called a pileup.

Listen for the pattern

Before joining, copy the station's call sign, its exchange, and its instructions. Does it send on one frequency and listen on another? Is it asking for a region or a particular partial call? Is it taking short reports or having longer contacts?

Send your complete call once, then listen. If you don't get a reply, wait for the next invitation. Repeatedly calling over the person who was selected slows the whole queue down.

A split example

Imagine a station transmitting on 14.025 MHz and sending UP 1. You listen on 14.025 and call on 14.026 MHz, provided the frequency is permitted and otherwise usable.

Your setting	Example
Receive	14.025 MHz: hear the station
Transmit	14.026 MHz: call where it listens
Operating mode	CW
Before keying	Check both frequencies and the split setting

These are illustrative frequencies, not a recommendation to occupy them. UP 1-3 announces a listening range one to three kHz higher. UP by itself doesn't state an exact offset; listen for clarification and for where the successful callers are being answered.

RIT usually changes only reception. XIT changes the transmit offset. SPLIT commonly assigns different tuning channels to receive and transmit. Confirm how your particular radio implements them before calling. If you can't set the required transmit frequency, wait for a suitable opportunity rather than calling on top of the station's own signal.

A brief exchange

A: `YOURCALL 599`
B: `R UR 579 TU`
A: `TU THEIRCALL UP`

That's enough when the operator is working a queue and has asked for that exchange. If your call was wrong, correct it before the contact ends. Save the station description for a more relaxed conversation.

Timing beats persistence

A readable signal in a listening gap has a better chance than a long call nobody can separate from the noise. Keep track of the operator's rhythm. Respect a reply to someone else. A short pause is useful operating time.

CHAPTER 17

Clubs, portable activities, and contests

Organized activities give you a reason to get on the air and people to look for. They also bring their own exchanges and equipment rules. Read those before treating a familiar-looking contact as an entry for an award.

Portable contacts

An operator on a hill, in a park, or at a temporary station may have a queue, a small battery, and changing weather. A short report and a friendly goodbye can be exactly right. If they invite a conversation, you're free to stay a little longer.

Some programs use a location reference. Copy it completely when you need it, including the prefix and punctuation used by the program. Ask REF AGN PSE if it disappeared in fading. A nearby landmark on a map isn't enough to invent the reference.

Different programs handle logs differently. Some rely on the activator's log, some allow separate participant submissions, and award rules vary. Your own accurate station log remains useful in every case.

Key-based clubs

A club may encourage straight keys, bugs, or sideswipers and set limits on electronically generated sending for its awards. The physical appearance of a paddle doesn't tell you whether the transmitted timing is manual. Check the actual key policy, including any accessibility provisions.

You can enjoy a conversation with a member without claiming to meet a club's award conditions. If asked for a membership number you don't have, say I AM NOT A MEMBER. Don't make one up.

Contests

A contest exchange might include a serial number, zone, section, age, region, or another field. There is no universal number that works for all events.

Here is an example only for an event that requires RST and a serial number:

A: `CQ TEST THEIRCALL`
B: `YOURCALL`
A: `YOURCALL 5NN 023`
B: `5NN 007`
A: `TU THEIRCALL TEST`

023 is A's number. 007 is yours. You don't echo their serial number back as your own. Ask NR AGN PSE if you missed it.

Before entering, check the operating period, allowed bands, category, exchange, repeat-contact rules, and report format. A logging program helps once it is set up for the right event. A paper log is still a useful fallback.

Keep it friendly

An event is a shared activity. Let the active operator set the pace. Ask briefly when you need a repeat, and give the requested information. If you want a slower practice contact, a club practice session or an arranged meeting may be a better fit than the busiest part of a contest.

CHAPTER 18

Logs and confirmations, on paper or online

A log is the record of what happened. A confirmation is a separate acknowledgment of that contact. Keeping those two ideas apart makes both simpler.

What to write during the contact

Field	Why it matters
UTC date and time	Identifies when the contact happened, including the date at midnight
Other call sign	The actual station, including relevant suffixes
Frequency or band	Where you worked
Mode	CW for these examples
Report sent	Your report of the other signal
Report received	The report they gave you
Name and location	Useful conversation details
Event exchange or reference	Fields required by the activity, when applicable

A pencil and ruled notebook work well. Leave separate boxes for sent and received reports. If you're busy, write the essential fields first and tidy the notes later.

Keep the direction straight

If you give 559 and receive 579, record **sent 559 / received 579**. The words sent and received are less ambiguous than two unlabeled numbers. They are also clearer than a lone S, which might mean either sent or signal strength.

Afterward

Check uncertain details while they are still fresh. A callbook or online lookup can help you spot a likely typo, but it can't establish a call sign you never copied. Keep a note of the uncertainty rather than silently filling the gap.

If you use logging software, export a backup in a common interchange format such as ADIF. Check a sample of the export for call signs, UTC dates, modes, frequencies, and reports. An export file isn't a confirmation service; it is a way to move your records.

Ways to confirm

A paper QSL card, an electronic matching service, or an event's log system may provide the acknowledgment you want. Choose what suits the activity. A confirmation accepted by one award program may not be accepted by another.

A service may compare records submitted independently by both stations. Your own upload can therefore remain unconfirmed until the other station contributes a matching record. Uploading the same entry everywhere doesn't guarantee every program will recognize it.

QSL during the contact often just means received. It isn't a promise that an online record has already matched. Ask about confirmation arrangements politely and follow the recipient's published instructions.

Shared and remote stations

Use the call sign and station identity authorized for the operation. A club station's records belong in the club's agreed logging process. If a receiver or transmitter was remote, record the relevant station location and check the activity's rules about remote assistance and receiving sites.

A remote receiver can be excellent for learning, but reception at that site should not be described as reception at your home antenna.

CHAPTER 19

Practice that fits your life

A fifteen-minute session

Time	Exercise
3 minutes	Familiar letters and digits, mixed
4 minutes	Short words or call signs
4 minutes	One small contact scenario
2 minutes	Send a short passage and listen back
2 minutes	Note the two things to revisit next time

That's a suggested routine, not a deadline. Regular sessions are easier to sustain when they end before you're exhausted. Keep an ordinary record of what improved and what still needs work.

An entirely offline route

Use an oscillator and a key. Ask a partner to send a few words from this book without telling you which ones. Record a short session if you can, so that both of you can listen again. Swap roles and change the names, reports, and numbers each time.

A receiver adds the challenge of real timing, noise, and unfamiliar operators. Try copying one call sign and a report before attempting a whole conversation. You can learn from a contact without being able to copy every word.

If you use a trainer

Look for adjustable character and effective speeds, random groups, a way to repeat a passage, and a clear record of errors. Browser, desktop, phone, and dedicated hardware trainers can all provide useful practice. Choose a tool you can use regularly rather than collecting several you rarely open.

If a program accepts text, start with a few phrases from the phrase bank. Listen with the text hidden, write what you heard, and compare afterward. For a joined prosign, use the program's documented prosign syntax. Typing separate A and R won't necessarily produce [AR].

If you use a decoder

Try listening first and checking the screen second. A decoder can expose a missed word or help you explore faster material. It can also teach you to trust an incorrect display if you never check it.

Adjust audio level and tuning before blaming your ears. Several signals at once, uneven spacing, and fading can all degrade the result. On air, ask for a repeat rather than treating an uncertain decoded field as confirmed.

Move from rehearsal to conversation

Arrange a short contact with an operator who knows your level. Agree on UTC, a starting frequency, and an alternative if it is occupied. Say what you can comfortably receive. Asking for a practice contact is more useful than promising a speed you can't yet follow.

You are ready for that arranged contact when you can send your call, copy basic numbers, ask for a repeat, request QRS, and recognize the closing. You don't need the entire dictionary memorized first.

CHAPTER 20

Use the web when it helps

The web can supply practice audio, a receiver in another location, a clue about who is active, or a place to keep a log. Those are different jobs. Decide which one you need before choosing a site.

A receiver in your browser

A web-accessible receiver takes its signal from an antenna at the receiver's own location. You select a supported frequency and mode, start the audio, and listen. Many also show a waterfall, which is a useful visual map of signals over time.

You don't need to attach your own radio or antenna to listen this way. You do need to remember where that receiving antenna is. A station heard there may be inaudible at home, and the reverse can also be true.

Use CW mode when available, or a suitable SSB setting according to that receiver's instructions. Begin with a usable filter width, find the tone, and refine the tuning. Some browsers require a click before audio will play.

A spotting network

A spot says that a person or an automatic receiver detected a station at a particular time and frequency. It is a lead to follow. It isn't the received audio, a guarantee of current activity, or proof of a two-way contact.

A practical sequence is: find a likely station, tune a receiver, listen to the instructions, and decide whether you can make the contact. The spot helps you find the conversation; your receiver tells you what is happening now.

Online practice and keyed chat

An online trainer can send text as audio. A networked Morse room may let participants exchange keyed timing. A key often needs an interface supported by that particular system; a plug that fits doesn't establish compatibility.

Internet practice can be social and useful, especially when you have no local partner. Label it honestly in your notes. A network exchange alone isn't an amateur RF contact or an automatic entry for a radio award.

Monitoring your own signal

With permission to transmit, you may be able to hear your signal through a remote receiver. Allow for network delay and differences in propagation. Delayed audio isn't a good replacement for a local sidetone when you're trying to time your hand movements.

An automatic spot can also provide feedback that a receiving node detected your call. No spot doesn't prove that the transmitter was silent: reception, coverage, recognition, and network operation all play a part.

Optional examples

The reference section lists a few training, receiver, spotting, and logging resources. They are starting points, not required accounts or endorsements. A notebook, a receiver, and a practice partner remain a complete way to work through this guide.

CHAPTER 21

One hundred words to hear as words

Use these words as listening material or as plain vocabulary during a contact. They aren't an extra set of procedure codes. Spaces in the pattern separate letters. For practice, hide the pattern and listen to the word before checking it.

Words 1-20

Word	Morse pattern
NAME	-. .- -- .
CALL	-.-. .- .-.. .-..
STATION	... - .- - .. --- -.
OPERATOR	--- .--. . .-. .- - --- .-.
CITY	-.-. .. - -.--
COUNTRY	-.-. --- ..- -. - .-. -.--
HOME	.... --- -- .
HERE	.... . .-. .
NEAR	-. . .- .-.
FROM	..-. .-. --- --
HELLO	.... . .-.. .-.. ---
MORNING	-- --- .-. -. .. -. --.
AFTERNOON	.- ..-. - . .-. -. --- --- -.
EVENING	. ...- . -. .. -. --.

Word	Morse pattern
NIGHT	-. .. --. .... -
THANKS	- .... .- -. -.- ...
PLEASE	.--. .-.. . .- ... .
SORRY	... --- .-. .-. -.--
GOOD	--. --- --- -..
LUCK	.-.. ..- -.-. -.-

Words 21-40

Word	Morse pattern
AGAIN	.- --. .- .. -.
REPEAT	.-. . .--. . .- -
SLOW	... .-.. --- .--
FASTER	..-. .- ... - . .-.
WAIT	.-- .- .. -
READY	.-. . .- -.. -.--
HEAR	.... . .- .-.
COPY	-.-. --- .--. -.--
READ	.-. . .- -..
SEND	... . -. -..
RECEIVE	.-. . -.-. . .. ...- .
SIGNAL	... .. --. -. .- .-..
REPORT	.-. . .--. --- .-. -
STRONG	... - .-. --- -. --.

Word	Morse pattern
WEAK	.-- . .- -.-
CLEAR	-.-. .-.. . .- .-.
NOISE	-. --- .. ... .
STATIC	... - .- - .. -.-.
FADING	..-. .- -.. .. -. --.
FILTER	..-. .. .-.. - . .-.

Words 41-60

Word	Morse pattern
RADIO	.-. .- -.. .. ---
KEY	-.- . -.--
PADDLE	.--. .- -.. -.. .-.. .
ANTENNA	.- -. - . -. -. .-
DIPOLE	-.. .. .--. --- .-.. .
VERTICAL	...- . .-. - .. -.-. .- .-..
WIRE	.-- .. .-. .
POWER	.--. --- .-- . .-.
WATTS	.-- .- - - ...
BATTERY	-... .- - - . .-. -.--
GROUND	--. .-. --- ..- -. -..
FREQUENCY	..-. .-. . --.- ..- . -. -.-. -.--
BAND	-... .- -. -..
TUNE	- ..- -. .

Word	Morse pattern
ABOVE	.- -... --- ...- .
BELOW	-... . .-.. --- .--
UP	..- .--.
DOWN	-.. --- .-- -.
CHANGE	-.-. .... .- -. --. .
LISTEN	.-.. .. ... - . -.

Words 61-80

Word	Morse pattern
CONTACT	-.-. --- -. - .- -.-. -
FRIEND	..-. .-. .. . -. -..
FIRST	..-. .. .-. ... -
LAST	.-.. .- ... -
NEW	-. . .--
OLD	--- .-.. -..
TODAY	- --- -.. .- -.--
TOMORROW	- --- -- --- .-. .-. --- .--
LATER	.-.. .- - . .-.
TIME	- .. -- .
MINUTE	-- .. -. ..- - .
HOUR	.... --- ..- .-.
DAY	-.. .- -.--
WEEK	.-- . . -.-

Word	Morse pattern
YEAR	-.-- . .- .-.
SOON	... --- --- -.
WEATHER	.-- . .- - .... . .-.
SUNNY	... ..- -. -. -.--
CLOUDY	-.-. .-.. --- ..- -.. -.--
RAIN	.-. .- .. -.

Words 81-100

Word	Morse pattern
SNOW	... -. --- .--
WIND	.-- .. -. -..
WARM	.-- .- .-. --
COLD	-.-. --- .-.. -..
PLUS	.--. .-.. ..- ...
MINUS	-- .. -. ..- ...
NORTH	-. --- .-. - ....
SOUTH	... --- ..- - ....
EAST	. .- ... -
WEST	.-- . ... -
PORTABLE	.--. --- .-. - .- -... .-.. .
MADE	-- .- -.. .
BUILD	-... ..- .. .-.. -..
PRACTICE	.--. .-. .- -.-. - .. -.-. .

Word	Morse pattern
LEARN	.-.. . .- .-. -.
ENJOY	. -. .--- --- -.--
BEGINNER	-... . --. .. -. -. . .-.
MEET	-- . . -
NEXT	-. . -..- -
REPLY	.-. . .--. .-.. -.--

CHAPTER 22

Small exercises, with answers

Pick your response

1. Someone asks QRL? while you are already using the frequency.
2. You received the report but not the name.
3. Your correspondent is sending faster than you can copy.
4. They call another station whose suffix resembles yours.
5. They copied your call with the wrong final letter.
6. The station says UP 1 and your transmit and receive frequencies are identical.
7. You gave 559 and received 579. Which number goes in each log field?
8. A decoder suggests a call sign, but you're unsure about the digit.
9. The agreed frequency for your sked is busy.
10. A station asks for a club number you don't have.

Answers

1. Briefly indicate that the frequency is occupied: QRL or C is sufficient.
2. RST CPY, then NAME AGN PSE.
3. PSE QRS; add a comfortable speed if helpful.
4. Keep listening until the other station has finished.
5. Repeat your whole call sign clearly.

6. Set and check the appropriate permitted transmit frequency before calling.
7. Sent 559; received 579.
8. Ask for the call again.
9. Use a clear alternative; the arrangement wasn't a reservation.
10. Say I AM NOT A MEMBER.

Read this block

```
R TNX FER RPT [BT] HR PWR 10 W ANT DIPOLE [BT] WX RAIN
TEMP PLUS 12 C [BT] PSE QRS [KN]
```

The sender received your report, uses ten watts and a dipole, has rain and a temperature of twelve Celsius, and would like you to slow down. The reply is directed to the named station. The block doesn't give a new report of your signal or a location. Don't add either to the log by inference.

Repair this call

```
CQ QRZ QRL DE YOURCALL 599 SK
```

These groups do different jobs. Listen and ask QRL? separately. When the frequency is clear, call CQ CQ DE YOURCALL YOURCALL K. QRZ? asks who called you; a report follows contact with an identified station; [SK] closes a contact.

Make the rehearsal unpredictable

Take a short exchange and change the name, town, report, and power each time. Ask a partner to alter one field without warning. You'll find out whether you are receiving the information or replaying a memorized script.

CHAPTER 23

Sources, sharing, and corrections

This English community edition was rewritten for a general audience. The examples, explanations, phrase selections, and layout were prepared for this guide. The Morse patterns and established operating conventions are shared technical facts; they are not inventions of the author.

Technical references

- **ITU-R M.1677-1, International Morse code.** The reference for international characters and standard timing. Document page[1].
- **ITU-R M.1172, miscellaneous radio abbreviations and signals.** A formal Q-code reference for the maritime mobile service. The guide distinguishes its formal meanings from conversational amateur usage. Document page[2].
- **CWops, CW Abbreviations.** A cross-check of common amateur shorthand. Reference list[3].
- **ARRL, Quick Reference Operating Aids.** A cross-check for reports and operating signals. Reference page[4].

These are references, not endorsements. Their documents and websites keep their own rights and licenses. No pages, illustrations, or substantial passages from third-party books are reproduced here.

Further reading

Dan Romanchik's *The CW Geek's Guide to Having Fun with Morse Code* offers another operator's approach to learning and operating. Roger Cooke's *Morse Code for Radio Amateurs* and Rob Brownstein and Jim Talens's *Morse Code Operating for Amateur Radio* are further options. Look for authorized copies from the authors or publishers. These books are not part of this guide's open license.

IARU's operating guidance is another useful perspective. Consult the current edition and its own license if you want to reuse its text. This guide is not an official IARU publication or a translation of that guidance.

Optional web resources

Job	Examples to explore
Training	LCWO; CWops resource directory
Remote listening	WebSDR directory; other web receiver directories
Automatic spots	Reverse Beacon Network
Logs and confirmations	A local ADIF-capable logger; QRZ Logbook; LoTW; eQSL; Club Log

Try the category you need. You don't need all of these services, and a service's support, accounts, rules, and award acceptance can change. No particular site is required for the exercises or contacts in this book.

LCWO[5] · CWops resources[6] · WebSDR[7] · Reverse Beacon Network[8] · QRZ Logbook[9] · LoTW[10] · eQSL[11] · Club Log[12].

References checked for this edition: 29 October 2026. Online material can change after publication. Check current national

rules, band plans, club requirements, and event regulations before relying on a specific permission or award condition.

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Corrections are welcome. A useful report includes the edition, chapter or page, the wording at issue, a proposed correction, and a reliable source where one is needed. A concrete example from practice can help distinguish a regional habit from a universal rule.

AI tools were used in drafting, research support, typesetting, and consistency checks. This is an independently prepared community guide, not a claim of certification or external peer review. Read critically, compare sources, and help improve the next edition.

CHAPTER 24

The desk card

Before you key

Listen. Copy the complete call. Check the mode and transmit frequency. Send at a speed you can receive.

Find room or answer

`QRL?` — Is this in use? Listen before calling.
`CQ CQ DE YOURCALL YOURCALL K` — A short CQ.
`THEIRCALL DE YOURCALL K` — A clear answer.

Give a first report

```
THEIRCALL DE YOURCALL GE [BT] UR RST 559 [BT] NAME ALEX
[BT] QTH RIVERDALE [BT] HW? K
```

Replace the report, name, town, and call signs with the real information.

Get back on track

`UR CALL AGN PSE` — Repeat your call.
`NAME AGN PSE` — Repeat your name.
`MY RST AGN PSE` — Repeat the report you gave me.
`PSE QRS` — Send more slowly.
`[AS]` — Wait.

Close and log

```
TNX QSO 73 THEIRCALL DE YOURCALL [SK]
```

UTC date · UTC time · call sign · band or frequency · CW · report sent · report received.

Leave space for the reply. Ask for what you missed. Enjoy the contact.

Reference links

The numbered links in the text refer to this list. Links remain active in the PDF. You can use the guide and its exercises without opening any of them.

[1] https://www.itu.int/rec/R-REC-M.1677-1-200910-I/en
[2] https://www.itu.int/rec/R-REC-M.1172-0-199510-I/en
[3] https://cwops.org/wp-content/uploads/2023/09/CW-Abbreviations.pdf
[4] https://www.arrl.org/quick-reference-operating-aids
[5] https://lcwo.net/
[6] https://cwops.org/cw-resources/
[7] https://www.websdr.org/
[8] https://www.reversebeacon.net/
[9] https://logbook.qrz.com/
[10] https://lotw.arrl.org/
[11] https://www.eqsl.cc/
[12] https://clublog.org/
[13] https://creativecommons.org/licenses/by/4.0/

Codes, phrases, and operating procedures.

Learn the rhythm, make a first contact, and keep useful words within reach. This guide is written for operators who enjoy a hand key and a notebook, those who use web tools, and everyone in between.

Inside: 139 abbreviation entries, 38 Q signals, 91 sample phrases, a worked contact, and a one-page desk card.

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